

```
import csv
```

```
file_path = "weather.csv" # uploading dataset in Colab
```

```
def map_snowfall(file_path):  
    mapped = []  
  
    with open(file_path, 'r') as file:  
        reader = csv.reader(file)  
        next(reader) # skip header  
  
        for row in reader:  
            station_id = row[0]  
            date = row[1]  
            element = row[2]  
            value = row[3]  
  
            # Filter for 2013 and snowfall  
            if element == "SNOW" and date.startswith("2013"):  
                mapped.append((station_id, date, int(value)))  
  
    return mapped
```

```
def reduce_max_snowfall(mapped_data):  
    max_record = None  
    max_snow = -1  
  
    for record in mapped_data:  
        station_id, date, snow = record  
  
        if snow > max_snow:  
            max_snow = snow  
            max_record = record  
  
    return max_record
```

```
with open("weather.csv", "w") as f:  
    f.write("station_id,date,element,value\n")  
    f.write("USW0001,20130101,SNOW,5\n")  
    f.write("USW0002,20130201,SNOW,12\n")  
    f.write("USW0003,20130115,TMAX,30\n")  
    f.write("USW0004,20130310,SNOW,20\n")
```

```
mapped_data = map_snowfall(file_path)  
result = reduce_max_snowfall(mapped_data)
```

```
if result:
    station_id, date, snow = result

    print("Maximum Snowfall Record:")
    print("Station:", station_id)
    print("Date:", date)
    print("Snowfall:", snow)
else:
    print("No snowfall data found for 2013.")
```

```
Maximum Snowfall Record:
Station: USW0004
Date: 20130310
Snowfall: 20
```

```
with open("weather.csv", "w") as f:
    f.write("station_id,date,element,value\n")
    f.write("USW0001,20130101,SNOW,5\n")
    f.write("USW0002,20130201,SNOW,12\n")
    f.write("USW0003,20130115,TMAX,30\n")
    f.write("USW0004,20130310,SNOW,20\n")
```